

Who Pays for the KV Cache?

Attributing Shared AI Inference Spend Across Kubernetes and LLM Provider Bills

Timothy Urista
Independent Researcher

github.com/timurista/unalloc · doi:10.5281/zenodo.22761013 · September 2026

Abstract. Organizations pay for AI through disconnected ledgers: Kubernetes allocations for self-hosted inference, gateway logs, and per-token bills from API providers. We present **unalloc**, an open-source tool that joins OpenCost, LiteLLM, OpenAI and Anthropic cost data into one Decimal-exact ledger and reports the share of spend with no owner, and use it to study where attribution breaks at the seams between these systems. Five case studies run inference for real or simulate it: a vLLM-style serving simulator with paged KV memory and prefix caching; a PyTorch transformer serving a multi-tenant trace with a real KV cache; tensor- and pipeline-parallel inference on **torch.distributed**; the unmodified CLI against mock provider APIs; and four downstream use cases. At the seams, owner labels set only on LeaderWorkerSet leader pods leave **66%** of a distributed deployment’s GPU bill unowned, and the natural fallback key assigns **61%** of it to a Helm chart name while the headline unallocated share falls to 4%; enabling every source double counts all gateway spend; and reading one page of a billing API reports a quarter of spend. Inside a shared inference server the metering rule decides who pays: on an NVIDIA H100 running vLLM, a token meter assigns a retrieval-heavy tenant **12–14 percentage points** more of the bill than an equal time-share meter at every load tested, while GPU utilization reads 97–99% from 2 to 16 requests per second and power draw tracks load. Neither meter is a ground truth, and we position these results against recent Shapley-based energy attribution. Code, raw data, captured evidence, figures and this paper regenerate from the repository.

1 Introduction

The cost of an AI feature is split across systems that were never designed to agree. A retrieval-augmented answer might touch a vector database and an embedding job on Kubernetes, a self-hosted open-weights model on a GPU pool, and a frontier model behind an API gateway. OpenCost [11] can price the first two; the gateway and the provider’s billing API can price the last. None of them can say what the feature costs, and none can say what share of the month’s spend belongs to no one — the number FinOps allocation practice is built around [14].

unalloc exists to produce that number. Its design is deliberately small (§2): every source is parsed into one **CostRow** type, labels are canonicalized so **label_costCenter**, **team_id** and a **team:platform** request tag join, and attribution groups the joined ledger on a chosen owner key. The interesting questions are empirical. Shared inference servers batch requests from many tenants [2], share KV memory through paged allocation and prefix caching [1, 4], and scale across devices with tensor and pipeline parallelism [5, 6]. Each of these mechanisms moves cost between tenants, and each leaves a trace — or fails to — in the ledgers a tool like **unalloc** reads.

We ask four questions, each answered by a case study:

1. **Metering.** When one inference pod serves several tenants, how much does the choice of meter (tokens, list-price tokens, compute time, KV memory) change each tenant’s bill? (§4, §5, §9)
2. **Distribution.** What happens to attribution when one model replica spans several pods? (§6)
3. **Joining.** What goes wrong when gateway and provider ledgers are combined? (§7)
4. **Use.** What can an organization do with a joined ledger beyond a percentage? (§8)

Contributions. (i) An open-source tool that joins Kubernetes, gateway and provider cost ledgers, with exact money handling, deterministic label canonicalization, invoice reconciliation, fallback-aware reporting

and a CI budget gate. (ii) Attribution failures at the seams between those ledgers, measured end to end: label propagation in multi-pod serving, fallback misattribution, double counting and truncated billing reads (§6, §7). (iii) A comparison of metering rules for shared inference servers in simulation, with real PyTorch inference on CPU, and with vLLM on an H100 (§4, §5, §9), positioned against work that measures Shapley ground truth for energy (§10). (iv) A reproducible artifact: raw data, the captured GPU bring-up and teardown, and one-command regeneration. Defects the studies exposed in the tool itself, each now covered by a regression test, are listed in Appendix A.

2 The joined ledger

2.1 Model

A **CostRow** is one unit of spend: a source, a cost-object name, an amount in USD as a **Decimal**, a time window, normalized labels, and an optional quantity (tokens, GPU-hours). Adapters only parse; they never aggregate or decide attribution. Given rows R , an owner key d and an ordered fallback list $f_1, \dots, f_k$, attribution assigns each row the first non-empty value of $d, f_1, \dots, f_k$, or **<unallocated>**:

$$U(R, d) = \frac{\sum_{r \in R, \text{owner}(r)=\perp} \text{amount}(r)}{\sum_{r \in R} \text{amount}(r)}$$

Because fallbacks make the headline smaller without making it more correct, **unalloc** also reports F , the spend attributed **only** through a fallback key (§6 shows why this matters).

2.2 Canonicalization

Label keys are lower-cased, camelCase is split, path-style keys keep their last segment (**app.kubernetes.io/name** $\rightarrow$ **name**), known provider prefixes are stripped until none remain, and an alias table maps spellings (**team_id**, **owner**, **squad**) onto one dimension. When distinct raw keys collapse onto one canonical key, a fixed precedence decides: a key already written canonically, then an alias, then a rewritten key, then a path-derived key, with ties broken on the raw key. Before the case studies the tool kept whichever key arrived first, which made results depend on provider serialization order.

2.3 Reconciliation and gates

reconcile compares each source’s ledger total with a billed amount and flags deltas above a tolerance; **report --budget P** exits non-zero when $U > P$, so a deployment pipeline can refuse to ship unowned spend.

3 Methodology

Table 1 summarizes the studies. Every study emits payloads in the providers’ own response shapes and parses them with **unalloc**’s production adapters; the end-to-end study goes further and runs the CLI binary against HTTP servers. Prices are deliberately round: one H100-class GPU at \$3.00/hour, OpenCost’s default CPU and RAM rates, and illustrative per-million-token API prices. The questions are about **shares**, which do not depend on the absolute rates. Unless stated, a monthly pool is $8 \text{ GPUs} \times 720 \text{ h} = \$17,280$.

Study	What runs	Real vs. modelled	Runtime
kv_cache	Discrete-event vLLM-style engine: paged KV blocks, prefix caching, chunked prefill, continuous batching, recompute preemption; 4 tenants, Poisson + multi-turn sessions	modelled latency	102 s
torch_kv	3.3M-parameter decoder (4 layers, $d = 256$, RoPE [9]) with explicit KV cache; 96-request 4-tenant trace on CPU	real inference	29 s
distributed	Megatron-style TP (2, 4 ranks) and 2-stage PP on torch.distributed /gloo; LeaderWorkerSet cost scenarios	real inference	24 s
hybrid_e2e	CLI subprocess vs. mock OpenCost, LiteLLM, OpenAI, Anthropic APIs with real auth schemes and cursor pagination	real CLI, synthetic spend	2 s
use_cases	Labeling Pareto, feature unit economics, self-host break-even, CI gate	mixed	25 s
gpu_validation	vLLM 0.29.0 serving Qwen2.5-7B-Instruct on one NVIDIA H100 80GB, driven with the kv_cache tenants at 2–16 requests/s (§9)	real serving, synthetic traffic	9 min

Table 1: Case studies. Runtimes on an Apple-silicon laptop (10 cores), Python 3.11, PyTorch 2.14 CPU, except **gpu_validation**, which ran on a DigitalOcean GPU Droplet.

4 A shared vLLM pod: the meter decides who pays

4.1 Setup

The simulator models one replica of an 8-replica fleet serving an 8B-parameter model (131 KB of KV per token) from a 12,000-block pool (16 tokens per block). Engine steps share a 2,048-token budget between decodes and chunked prefills [3], and step latency is an analytic function of prefill tokens, decode sequences and attended context. Four tenants arrive by Poisson process: **search** (RAG: 600–1,400-token prompts, 16–64-token answers, three shared system prompts), **agents** (multi-turn sessions that re-send the whole conversation each turn, 150–400-token outputs), **platform** (medium/medium) and an unlabeled shared **sandbox** key. Each step’s duration is split across the sequences in it by their marginal work; each KV block-second is split across the requests holding the block. Idle time and unheld blocks are recorded separately as overhead.

At 3 requests/s for 30 minutes the engine served 10,020 requests with a 75% prefix-cache hit rate, no preemptions, and a median time-to-first-token of 38–96 ms per tenant. (At 4 requests/s a five-minute run looks healthy but agent follow-up turns accumulate over half an hour and median TTFT exceeds 40 s; we report the sustainable operating point.)

4.2 Results

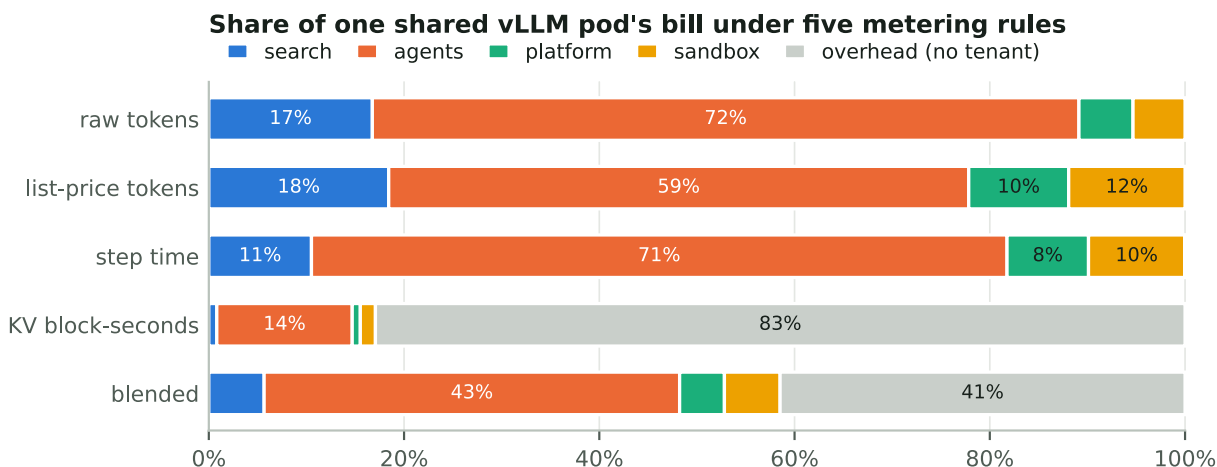


Figure 1: One pod bill, five meters. Token meters have no notion of overhead. Step-time metering sees none either once continuous batching keeps any request in flight; KV-memory metering leaves 83% of the bill with no request.

Figure 1 shows the split. Agents dominate every compute-like meter because each turn re-sends a growing conversation, but the meters still disagree: raw tokens charge **search** 16.7% of the pod, step time 10.5% — a 6.2-point gap that grows to 12.0 points against KV memory once overhead is redistributed. List-price weighting (cached input at 0.1 $\times$, output at 4 $\times$) is no closer: it moves agents from 72.4% to 59.4%, twelve points away from measured step time (71.2%).

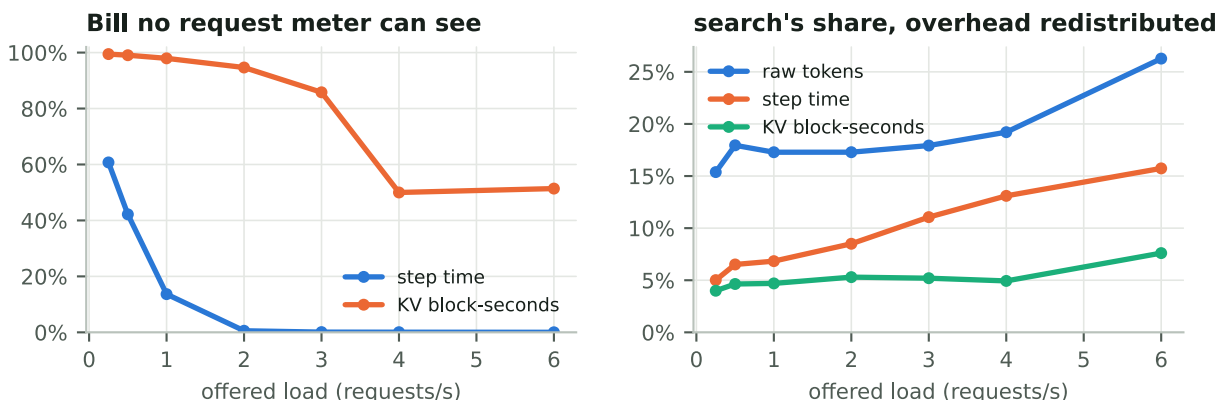


Figure 2: Left: overhead visible to each meter as load rises. Right: search's share under three meters, overhead redistributed.

Figure 2 explains why. Step-time overhead collapses to ≈ 0 by 2 requests/s: a pod that is “busy” in wall-clock terms is nowhere near its token capacity (754 vs. $\approx 1,340$ output tokens/s). KV memory tells the opposite story — 95% of block-seconds are unheld at the same load, because pools are provisioned for peaks. Neither is wrong; they measure different scarce resources, and the choice moves several points of the bill between tenants. §9 checks both observations against a real vLLM server on an H100.

For unalloc. A per-tenant split only closes the gap if overhead is labeled. Emitting the memory split with an unlabeled overhead row reports 84.5% unallocated; redistributing the same overhead as OpenCost **sharedCost** reports 9.1% — only the sandbox key. The unallocated percentage is as much a statement about the overhead policy as about labels.

5 Real inference with a KV cache

5.1 Setup

A from-scratch decoder-only transformer (3.28M parameters, random weights with a fixed seed — serving cost does not depend on weight values) implements a preallocated per-layer KV cache and shared-prefix reuse. Before any timing, cached decoding is checked against full recomputation: generated tokens are identical and the maximum logit difference is 1.07×10^{-6} . The trace has 96 requests from the same four tenant shapes; it is served three times and per-request medians are used. Measured shares moved by at most 4.3 points across repeats.

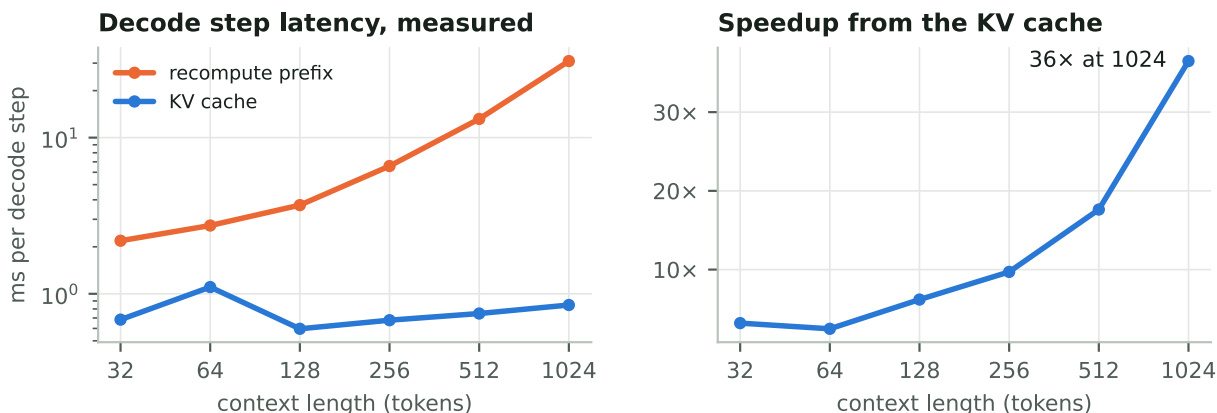


Figure 3: Measured decode-step latency on CPU. At 1,024 tokens of context the cache is 8.0 MiB and a step takes 0.85 ms instead of 30.9 ms.

5.2 Results

Figure 3 reproduces the expected scaling [7]: recomputation grows with context, cached steps stay flat, $36.5\times$ faster at 1,024 tokens. The attribution consequences are in Figure 4. Search sent 18,024 prompt tokens and received 701; agents sent 2,345 and received 4,017. Token counting (and analytic FLOPs [8], which track tokens) charge search $\approx 45\%$ of the pool. Measured compute charges it 12.4%: prefill is batched and cheap, decode is sequential and expensive.

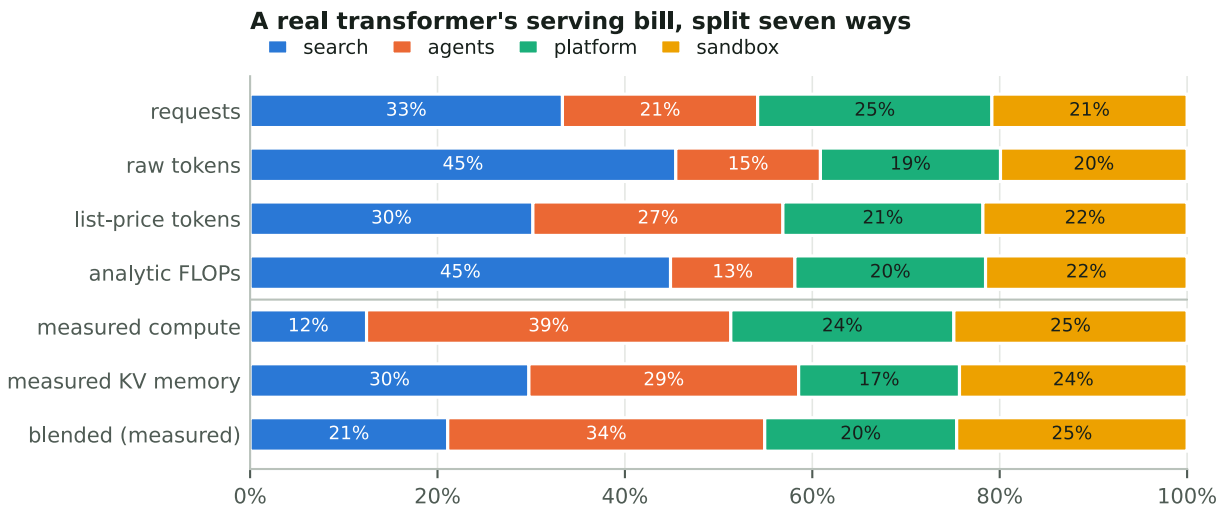


Figure 4: The same measured trace split seven ways. Above the rule: meters derivable from request logs. Below: meters that require serving telemetry.

Headline. Token counting and measured compute disagree on the RAG tenant’s share by **33.0 points** of the pool, \$5,703 of \$17,280 a month. Measured cost per token differs 9.2× between search and agents, which a flat token price charges identically; weighting output at 4× halves the disagreement (17.8 points) without closing it. Neither meter is a ground truth, and on a batching GPU server the disagreement is smaller (§9).

On a GPU the per-step fixed overhead is amortized across a batch, so the magnitude is hardware-specific; the direction — prompt-heavy tenants subsidize decode-heavy ones under token pricing — is structural, and matches the prefill/decode asymmetry that motivates disaggregated serving [10].

6 Distributed inference: attribution leaks through pod templates

6.1 Parallel inference, verified

We shard the reference model Megatron-style [5] — column-parallel QKV and MLP up-projection, row-parallel output and down-projection with an all-reduce after each — and separately split it into two pipeline stages that pass activations with send/recv [6]. All configurations generate the same tokens as the single-process model (maximum logit difference 3.3×10^{-6} for TP, exactly 0 for PP). Per-rank parameters drop from 13.65 MB to 7.36 MB (TP2) and 4.21 MB (TP4); embeddings, norms and the LM head stay replicated. KV cache per rank halves with each doubling.

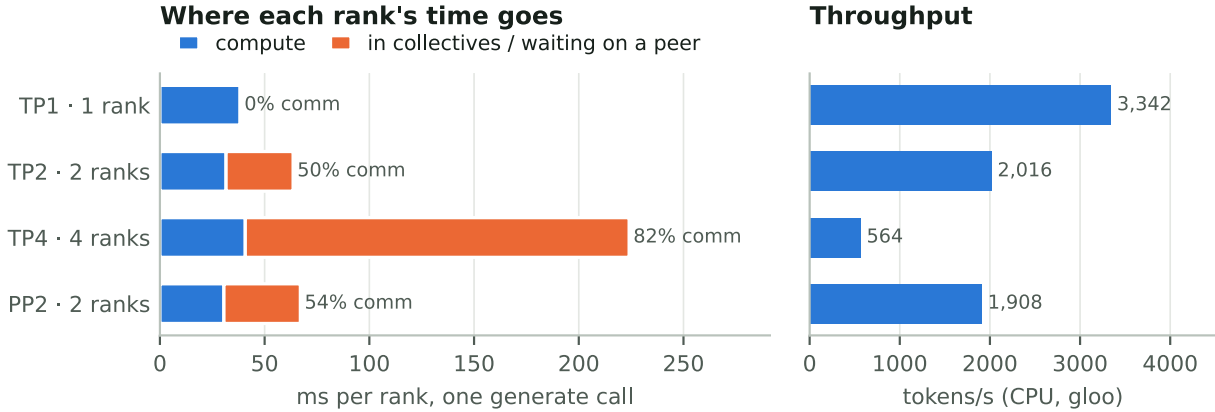


Figure 5: Per-rank time split for batch 4, 64-token prompt, 32 new tokens. Communication is time blocked in collectives or waiting for a peer.

On CPU over loopback, distribution is slower, as expected (Figure 5): TP4 spends 82% of each rank’s time in collectives. These fractions are an upper bound for NVLink-connected GPUs; we use them only to size the scenario below and report a sensitivity series at 5–20%.

6.2 The LeaderWorkerSet month

Multi-host serving on Kubernetes commonly runs one replica as a group of pods, e.g. with LeaderWorkerSet [13]: a leader template and a worker template. We emit one month of OpenCost allocations for two tenants — **search** (2 replicas × 4 pods, tensor parallel) and **agents** (3 replicas × 2 pods, pipeline parallel) — plus cluster idle, \$38,400 in total, with each pod’s GPU time split into compute, communication and in-pod idle using the measured fractions.

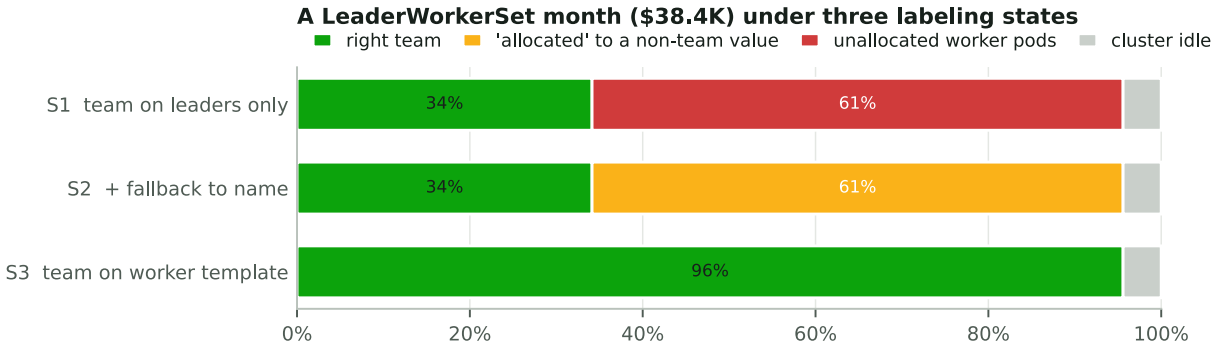


Figure 6: Correctness of attribution, not just coverage, under three labeling states.

In S1 the **team** label is on the leader template only: 65.9% of the month is unallocated. The obvious rescue is a fallback key every pod carries. In S2, falling back to **name** drops the unallocated share to 4.4% — and routes \$23,597 (61%) into a bucket called `vllm`, the Helm chart’s `app.kubernetes.io/name`, which collides with `leaderworker.set.sigs.k8s.io/name` on the canonical key **name**. S3 fixes the worker template: 95.6% correct, only idle remains.

For unalloc. The headline percentage cannot distinguish S2 from S3. We added the fallback-attributed amount F to every report (\$23,597 in S2, \$0 in S3) and made collision resolution order-independent; before the fix the S2 bucket was `vllm` or `search-llama-70b` depending on JSON key order.

Communication is itself a cost with an owner. GPU time spent in collectives totals \$12,059, \$8,201 of it on worker pods that S1 leaves unowned. A per-token showback, which spreads it by token share, moves \$3,737 from the tensor-parallel tenant onto the pipeline-parallel one; at a more GPU-like 10% communication fraction the transfer is \$166.

7 Joining gateway and provider ledgers

The organization in this study runs a LiteLLM gateway [12] in front of OpenAI and Anthropic, the self-hosted fleet above, and — like most — a research team calling providers directly. Mock servers enforce each provider’s authentication (Bearer for OpenAI and LiteLLM, `x-api-key` for Anthropic’s Admin API) and page responses with cursors; in the dev container the LiteLLM spend log lives in Postgres, as it does in production. The CLI runs as a subprocess with only environment variables pointing it at the servers.

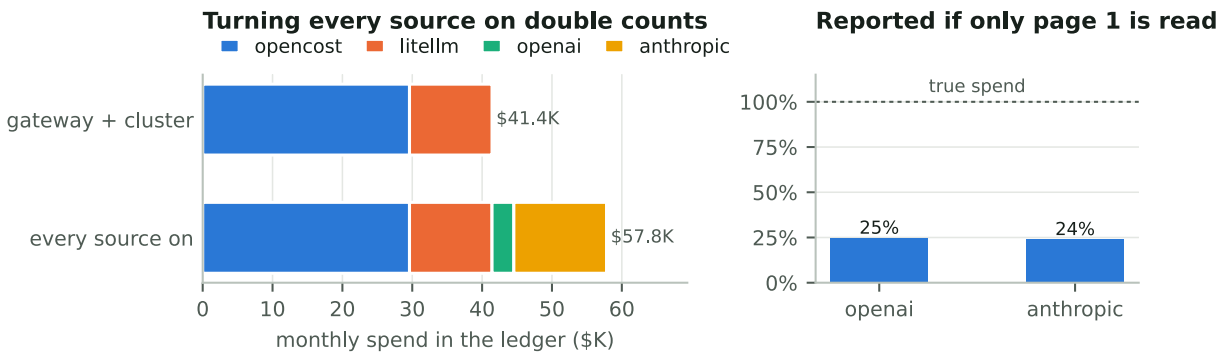


Figure 7: Left: ledger totals for the recommended sources versus every source. Right: share of true provider spend reported by an adapter that reads only the first page.

Scenario	Ledger total	Unallocated
A. Cluster + gateway (<code>--source opencost --source litellm</code>)	\$41,420	66.9%
B. Every source enabled	\$57,813	76.3%
C. Provider billing only	\$16,393	100.0%
C'. Provider billing, fallback to project/workspace	\$16,393	0.0%
Provider ledgers vs. provider invoices	matched	0.0% delta
Gateway ledger vs. provider invoices	\$11,815 of \$16,393	-27.9%

Table 2: End-to-end scenarios over one month.

Table 2 and Figure 7 give three results. **Double counting:** enabling every source adds \$16,393, of which \$11,815 — exactly the gateway’s spend — is counted twice; the remainder is real bypass traffic the gateway never saw. **Coverage:** the gateway ledger reconciles to only 72.1% of the provider invoices, so a gateway-only view under-reports by \$4,578 and cannot know it without reconciliation. **Fallback illusion:** provider billing alone has no team dimension; falling back to project and workspace reports 0% unallocated while attributing nothing to a team.

Before the fixes in Table 4, this study could not run at all: the OpenAI and Anthropic adapters ignored their default URLs whenever the CLI passed an unset environment variable, Anthropic rejected the Bearer header, and neither adapter followed pagination — reading 25% of OpenAI spend and 24% of Anthropic spend.

8 Use cases for a joined ledger

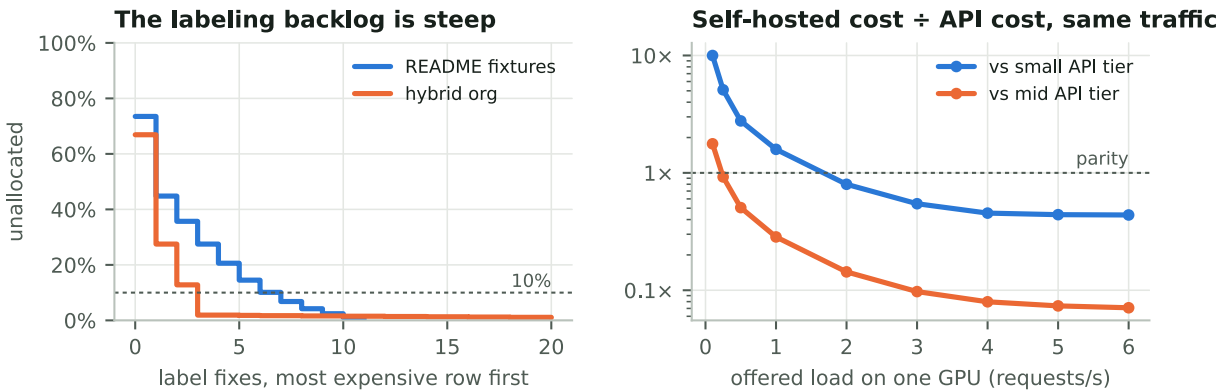


Figure 8: Left: unallocated share after fixing the most expensive unlabeled rows first. Right: cost of serving the simulated traffic on one self-hosted GPU divided by the same tokens at API list prices, for two illustrative price tiers.

Labeling is a Pareto problem. Ordering the backlog by dollars (Figure 8, left), three label changes take the hybrid organization from 66.9% to 1.9% unallocated; the README fixtures need seven changes to cross 10%. A dollar-ordered backlog turns a governance program into a short ticket list.

Feature unit economics need both ledgers. Joining on a **feature** label, the RAG **answer** feature costs \$22.32 per thousand requests including its vector database, against \$13.38 if only the LLM bill is counted: 40% of the feature’s cost is invisible to a gateway-only view.

Build-versus-buy depends on utilization, not list price. One self-hosted GPU serving the simulated traffic is cheaper than a mid-tier API above ≈ 0.23 requests/s and cheaper than a small-tier API above ≈ 1.7 requests/s (Figure 8, right), flattening at $0.08\times$ and $0.45\times$ respectively once saturated. The comparison bounds cost, not quality.

Budgets belong in CI. `unalloc report --budget 50` exits 2 on the fixtures (73.5% unallocated); `--budget 80` exits 0, so a deployment that ships unlabeled workloads can fail review.

9 Validation on a datacenter GPU

9.1 Setup

The CPU timings of §5 and the analytic latency of §4 leave one question open: does the metering result survive production serving software on production hardware? One DigitalOcean GPU Droplet (**gpu-h100x1-80gb**: NVIDIA H100 80GB HBM3, driver 580.173.02, CUDA 13.0, 20-vCPU Xeon Platinum 8468, 235 GB RAM) ran vLLM 0.29.0 [1] serving Qwen2.5-7B-Instruct in bf16 with an 8,192-token context and prefix caching; vLLM sized its KV cache at 995,296 tokens.

A client on the same machine replayed the §4 tenants as token-id prompts, so lengths were exact and shared system prompts byte-identical, with output lengths forced and greedy decoding. It ran two minutes each at 2, 4, 8 and 16 requests/s, recording each request’s time to first token, completion time and the server-reported cached-token count, and sampling vLLM’s Prometheus metrics twice a second and **nvidia-smi** once a second. The time-share meter splits every 50 ms of wall time equally across in-flight requests. The droplet existed for 29 minutes (about \$2.15); the runbook and the captured terminal output of every step, from creation to a verified deletion, are in the repository.

9.2 Results

Load	Requests	Output	Cache	TTFT	TPOT	GPU	Power	search	search
req/s		tok/s	hits	p50 / p95 ms	p50 ms	util	W	tokens	time
2	446	773	67%	28 / 44	6.3	97%	469	16.5%	4.8%
4	862	1,479	79%	26 / 43	6.7	99%	499	16.6%	4.7%
8	1,638	2,761	78%	29 / 47	7.4	99%	547	17.2%	4.6%
16	3,295	5,389	66%	51 / 99	12.2	99%	660	18.9%	5.2%

Table 3: vLLM on an H100. All 6,241 requests completed without error and none waited in the queue. TTFT is time to first token; TPOT is time per output token. The last two columns are search’s share of the bill under token and time-share meters, overhead redistributed.

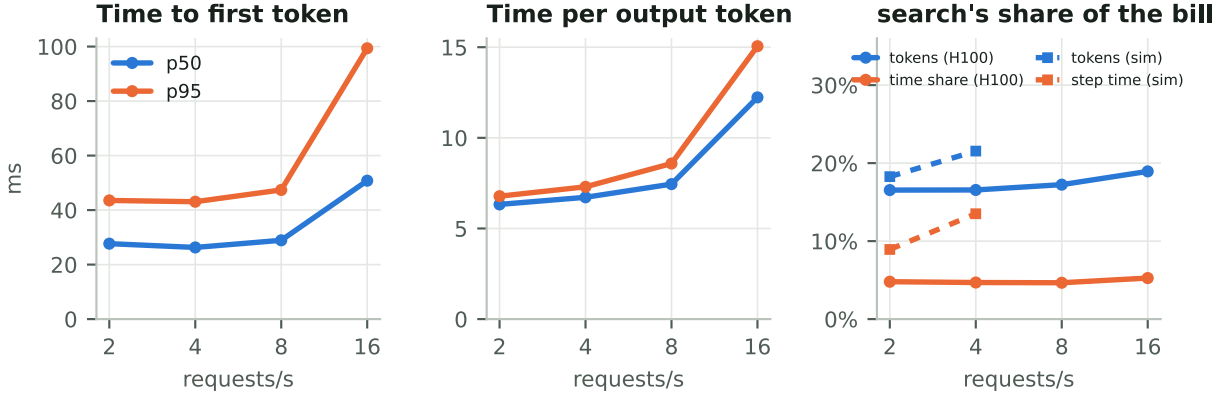


Figure 9: Left, middle: latency percentiles by load on the H100. Right: search’s share of the bill under token and time-share meters on the GPU (solid), and under the simulator’s token and step-time meters where the simulator is not saturated (dashed).

The meters still disagree. The token meter assigns search 16.5–18.9% of the bill; the time-share meter assigns 4.6–5.2%, a disagreement of 11.7–13.7 points at every load (Table 3, Figure 9). At 2 requests/s the simulator predicted 18.3% against 8.9%, a 9.4-point disagreement. The CPU experiment of §5 overstated the size, 33 points, because an unbatched decode step there carries large fixed overhead; on a batching GPU server the disagreement narrows but keeps its direction. Equal time share is itself a heuristic: it charges a request waiting on prefill the same as one decoding. A Shapley allocation measured by replaying tenant subsets, as JouleShare does for request energy [15], would provide a ground truth; with four tenants it needs 15 subset runs per load level.

Utilization is not a cost signal. `nvidia-smi` reported 97% utilization at 2 requests/s and 99% at 4, 8 and 16, while throughput rose 7 $\times$. vLLM had at least one running request in 97–100% of samples at every load, and the KV cache was 0.7–8.1% occupied. This is the §4 overhead result observed directly: time- and utilization-based meters see no idle capacity, and memory-based meters see almost nothing else. Power draw tracked load, from 469 W to 660 W.

The simulator’s latency was too pessimistic. Below saturation its throughput matched (737 vs. 773 output tokens/s at 2 requests/s), but its step-latency constants saturate it by 8 requests/s, with multi-second time to first token, while the H100 served 16 requests/s with a 99 ms p95. The served model also stores about 57 KB of KV per token against the simulator’s 131 KB. The §4 shares should be read as directional and its latencies as uncalibrated; fitting the simulator’s constants to this run is future work.

Operational. vLLM 0.29.0’s default FlashInfer sampler failed to JIT-compile on the provider’s GPU image and stopped the engine; the PyTorch sampler, equivalent for greedy decoding, worked. Deleting the droplet by tag cleared the tag before the droplet itself was gone, so teardown was verified by resource ID, not tag.

Further use cases. The same ledger supports multi-LoRA serving (attribution by adapter label on a shared base model), fine-tuning and batch embedding jobs whose GPU hours share clusters with online inference, agent systems whose tool calls fan out across providers, speculative decoding where a draft model’s cost belongs to the target model’s callers, and multi-cloud estates where Bedrock and Azure OpenAI bills join the same dimension.

10 Related work

Attributing shared inference. JouleShare [15] is the closest work to §4, §5 and §9. It measures exact request-level Shapley energy [19] on vLLM by replaying every subset of eight requests, finds that token-proportional attribution misallocates roughly a quarter of batch energy under both static and continuous batching, and fits a lightweight estimator to the measured Shapley shares. It is stronger than this paper on ground truth. We compare metering rules without one, at the level of tenants rather than requests, and include KV-memory, list-price and time-share rules alongside tokens. Vellaisamy et al. [16] decompose inference energy on H100 and H200 GPUs into fixed and per-token components and show that lower energy per token can reflect amortization rather than lower total energy: the same mismatch between token pricing and resource consumption that our meters express in dollars. LLMVisor [17] decomposes co-batched latency into per-request contributions for fractional sharing of a serving engine, again outperforming token-count baselines. PrefixShield [18] makes tenant groups responsible for the prefix-cache blocks they materialize, a scheduling answer to the shared-KV question our memory meter only measures. Shapley values [19] and dominant resource fairness [20] are the classical foundations for sharing joint costs and multi-resource capacity.

Cost and emissions allocation in clouds. Schneider and Mattia [21] allocate the energy and emissions of shared data-center machines, infrastructure and software to cloud users from machine-level measurements. ABACUS [22] is a FinOps service that sets budgets, blocks deployments that exceed them and predicts cost from infrastructure-as-code, the same gate `unalloc report --budget` applies to unallocated spend. Cost-Governed RAG [23] attributes per-tenant cost across embedding, retrieval and generation within one data platform, using a vector index whose memory is linear per tenant. Our feature economics (§8) join the same layers across systems that do not share a platform or a key: Kubernetes allocations [11], a gateway [12] and provider billing. FinOps allocation practice [14] supplies the organizational target all of these serve.

What this paper adds. Serving systems [1, 2, 3, 4, 10] define the mechanisms whose cost is attributed here but do not address billing, and the work above measures attribution inside one system. To our knowledge, none joins Kubernetes allocations, gateway logs and provider billing into one ledger or measures the attribution failures at those seams: label propagation across multi-pod serving templates, fallback keys that resolve to non-owners, double counting and truncated billing reads.

11 Discussion and limitations

Recommendations. (1) Decide the meter explicitly and publish it; token counts are a pricing choice, not a measurement. (2) Keep overhead — idle GPUs, empty KV pools, communication — as its own labeled line and redistribute it by policy, rather than letting any meter absorb it silently. (3) Put owner labels on every pod template of a multi-pod workload, and treat fallback-attributed spend as unverified. (4) Choose sources so each dollar has one path into the ledger, and reconcile against invoices every month.

Limitations. No allocation in this paper is a ground truth: the meters compared are defensible rules, and what we measure is how far they disagree, not which share is correct. The PyTorch timings of §5 and §6 are from CPU, where fixed per-step overhead and loopback collectives inflate decode and communication costs. §9 re-measures the metering result with vLLM on an H100, where its direction holds and its size is smaller. That run covers one GPU, one model, one two-minute window per load level and synthetic traffic, and its multi-turn prompts append synthetic assistant tokens, so cross-turn prefix reuse excludes previous answers. The serving simulator uses analytic step latency whose constants proved too pessimistic for that GPU. Dollar figures use round, illustrative prices. The mock provider APIs reproduce authentication and pagination semantics as documented, not every field of the live responses.

12 Reproducibility

Everything in this paper regenerates from the repository, archived as release 0.2.1 at [doi:10.5281/zenodo.22761013](https://doi.org/10.5281/zenodo.22761013):

```
make research          # CPU PyTorch, notebook tooling, typst
make case-studies      # writes case_studies/results/*/metrics.json
make paper             # figures + this PDF
make notebook          # executes the companion notebook
make devcontainer-check # the same, inside the isolated dev container
python -m case_studies.gpu_validation.analyze # §9, from the committed raw GPU data
```

The GPU run itself follows `case_studies/gpu_validation/RUNBOOK.md` on any single-GPU machine, after a free dry run against a bundled fake server. A browser-based ledger explorer (`python -m case_studies.ui`) shows every dataset row by row with the owner each row resolves to, and a companion notebook walks through each study.

Use of AI tools

The case-study code, experiment orchestration, analysis scripts, figures and drafts of this text were produced with Claude Code (Anthropic, Claude Opus 5) working under the author’s direction, including the GPU run in §9. Generative AI is not an author of this work; the author is responsible for all of its content.

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A Defects found by the case studies

The case studies exposed these defects in **unalloc** itself. Each is fixed and covered by a regression test; they are listed here rather than as findings because they concern the tool, not the attribution questions the paper studies.

Area	Defect	Found by
Fetch	OpenAI/Anthropic default base URL overridden by an unset env var; every live fetch failed	review; e2e
Fetch	Anthropic Admin API called with Bearer auth instead of x-api-key + anthropic-version	review; e2e
Fetch	No cursor pagination on provider cost APIs; first page only ($\approx 25\%$ of spend)	review; e2e
Money	OpenCost component costs summed through float when totalCost was absent	review
Labels	Only one provider prefix stripped; label_app_kubernetes_io_name $\neq$ app.kubernetes.io/name	distributed
Labels	Canonical-key collisions resolved by arrival order, incl. explicit key vs. alias	distributed, e2e
Labels	OpenCost annotations silently overrode same-key labels	distributed
Report	labels backlog ignored -- fallback , disagreeing with report	review
Report	labels printed “Nothing unallocated” when no source loaded	review

Table 4: Defects fixed in the course of the studies. Each has a regression test.